Ryan Crane

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SUMMARY

Senior mechanical engineering major and data science minor with foundational skills in circuit design, automation, and data analysis coupled with extensive experience in cell culture, synthetic biology, and project communication. Project work includes extensive projects in hardware design for experimental systems, with a publication and multiple conference presentations, and experience leading a research team (iGEM).

EDUCATION

Barrett, the Honors College, Arizona State University, Tempe, Arizona, USA

M.S. in Robotics and Autonomous Systems — Expected Fall 2028

B.S.E. in Mechanical Engineering – **GPA 3.88** — Expected Fall 2027

Relevant Coursework: Statics, Dynamics, Mechanics of Materials, Structural Mechanics, Principles of Mechanical Design, Thermodynamics, Fluid Mechanics, Heat Transfer, System Dynamics and Controls I and II, Computer Aided Design I (SolidWorks) and II (Ansys), Numerical Methods, Linear Algebra, Differential Equations, Physiology, Microbiology, Applied Experimental Statistics, Circuits

TECHNICAL SKILLS

CAD and Analysis: SolidWorks (50+ part assemblies), Ansys, FEniCS (FEA), MATLAB

Prototyping and fabrication: GD&T, FDM 3D Printing, Electromechanical Assembly, Soldering, Circuit Debugging

Motion and mechatronics: NEMA stepper motors, lead-screw linear stages, CNC XY stages, GRBL, robot-arm integration, Arduino, ESP32

Software: Python, MATLAB, C++, G-code, computer vision (image segmentation models)

Electronics: LTspice, Cadence Virtuoso, KLayout

Biology: Benchling, Cell Passaging, PCR, Gel Electrophoresis, Gram Staining, Quadrant Streaking, Serial Dilution, Bacterial Transformation, Gibson Assembly

WORK EXPERIENCE

Electrical Engineering Research Assistant — May 2026 - Present

NSF Athena AI Institute for Edge Computing, ***Duke University***

- Design and test CMOS synaptic integrated circuit to couple two silicon neurons (Leaky Integrate and Fire and Izhikevich) with current mirrors, operational transconductance amplifiers, and resistive non-volatile memory, minimizing area footprint in Cadence Virtuoso
- Analyze neuron spiking behaviors (RS, FS, tonic bursting, phasic spiking/bursting) and optimize tunable parameters through Monte Carlo simulations using Cadence Virtuoso
- Characterize transistor operating regions, analyze capacitance and frequency behavior with pico-scale analog design constraints
- Implement Izhikevich neuron circuit model and chip package routing with KLayout
- Design and run spiking neural network simulations in Python to test hardware-aware neural network synapse design against conventional simple mirror and cascode mirror designs that are prone to process variation, mismatch, and current error that propagates through the network
- Perform literature reviews to compare our proposed synaptic circuit design to existing approaches and prepare sections for our submission to a conference journal

Laboratory Automation Student Engineer — January 2025 - Present

BioProtean Lab, ***Arizona State University***

- Sole engineer on a vision-guided automation system for organoid culture: integrated a robotic arm, a cell-segmentation model, and motorized stages, and wrote the control code in Python and GRBL G-code
- Integrate a Seeed ReBot robotic arm to move a 96-well plate to and from an incubator
- Write control scripts in Python for the Seeed ReBot robotic arm
- Integrate computer vision through an organoid segmentation model capable of identifying a cell in a plate and moving the well to a safe position for pipetting
- Test automation system for use in the culture of organoids, 3D-stem cell derived tissue structures, to study the effects of human microglia and brain-derived cortical organoids
- Design and implement motorized focusing stage utilizing a lead screw and NEMA motor to translate rotational motion to translational motion of a stage holding an objective lens towards or away from the sample with micron precision
- Write scripts in GRBL G-code for control of autofocus routine

- Design a motorized well-scanning microscope with brightfield capability for organoid morphology analysis
- Integrate and write control scripts in CNC XY stage for a 96-well plate for use inside the Opentrons OT2 pipetting robot
- Write Python control scripts for the Opentrons OT2 pipetting robot to automate production of nanobodies through phage-assisted continuous evolution
- Operate Opentrons OT2 pipetting robot for experiments
- Troubleshoot, solder, and restore faulty laboratory equipment, including automated turbidostat system and surface plasmon resonance device
- Use oscilloscope, multimeter, power supplies for testing
- Design and 3D print microscope and automation system housing with 50+ individual components in SolidWorks
- Wrote step by step sequencing code for device integration and feedback for well adjustment based on captured images
- Conduct literature reviews and assemble documentation for conference publication and papers
- Budget for, find, order, and document all materials required for the automation system given an allocated budget
- Collaborate and coordinate with faculty, staff scientists, and graduate students across departments.

Synthetic Biology Hardware Team Lead November 2024 - November 2025
International Genetically Engineered Machine (iGEM) Team, ***Arizona State University***

- Conducted weekly meetings to assess the progress of the team and the hardware.
- Designed 3D printed casing and optimized electrochemical sensor for portability, usability, and reproducibility.
- Developed lectin-based electrochemical assay for antigen detection on red blood cells using a potentiostat circuit for cyclic voltammetry.
- Designed and optimized genetic constructs for glycoside hydrolase production in *Escherichia coli* and created a nitric oxide synthase-linked lectin to test a lectin-based assay using Benchling.
- Analyzed surface topography of SCOBY material using scanning electron microscopy (SEM).
- Analyzed leukocyte binding capability of SCOBY material using confocal fluorescence microscopy.
- Performed market analysis for available enzymatic conversion solutions on the market, developed an optimal timeline to bring our enzymatic conversion kit to market, and developed a business model.
- Interviewed multiple stakeholders and adjusted design, target market, and research & development plans of our kit from expert advice.
- Presented a poster and gave an oral presentation to 100+ established scientists, entrepreneurs, and students.

WORK EXPERIENCE CONTINUED

Synthetic Biology Hardware Team Engineer March 2024 - November 2024
International Genetically Engineered Machine (iGEM) Team, ***Arizona State University***

- First author publication in a hardware journal
- Designed and soldered circuits into a continuous bacterial cell culture system designed for phage assisted continuous evolution.
- Developed 3D printed housing and designed plumbing for positive pressure flow of media, bacteria, and virus.
- Wrote and optimized programs in Arduino IDE C++ to control the system's electronic components.
- Wrote feedback loops to keep turbidity steady
- Designed user interface for running separate protocols with the device
- Optimized turbidity data collection using a laser and photoresistor system and generated mid log growth curves.
- Applied molecular cloning techniques (PCR, gel electrophoresis, Gibson Assembly) to experimental workflows.
- Strengthened teamwork and communication skills in a fast-paced, interdisciplinary lab setting.

Ecology Research Assistant January 2023 – May 2023
Northern Arizona University

- Conducted experiments, collected and analyzed data, and presented findings to students, faculty, and professionals
- Gained hands-on research experience in aseptic technique and lab protocols, building foundational skills and familiarity with lab operations in a professional research setting.

HONORS AND AWARDS

Beckman Scholar 2025
- only 2 undergraduates selected at all of Arizona State University in 2025

iGEM Gold Medal 2024, 2025
- One of seventeen collegiate teams in the United States to win a gold medal

Dean's List 2023, 2024, 2025
- Must earn 12 or more graded credit hours with a GPA of 3.5 or higher
- Received in every semester at university

National Merit Scholar 2023

Flinn Scholarship Semi-Finalist 2023

PUBLICATIONS

Crane, R. D.*, Isaacs, W. T.*, Ijaz, M.*, Salazar, N., Botello, B., Kim, M., Flores-Mora F. E., Bartelle, B. B.. (2026) *The OptoPACE Bioreactor: An Accessible Turbidostat System with Phage-Assisted Continuous Evolution Capability and Optogenetic Controls*. Journal of Open Hardware 10.1

CONFERENCE PRESENTATIONS

Oral Presentations

Crane, R. D.*, Kim, M.*, Salazar, N.*, Cristobal, F.*, Scerca, SB*, Bartelle, B. B., et. al. (2025, October). *An Enzymatic Blood Conversion Kit for Use in Clinical Emergencies.* 2025 iGEM Grand Jamboree, Paris, France.

Crane, R. D.*, Hofstetter, R. (2023, May). *Not A Fun-Guy: Investigation of Climate Pollutants on the Survival of An Amphibian-Killing Fungus*. 2023 Senior Project Symposium, Flagstaff, Arizona, USA.

Poster Presentations

Crane, R. D.*, Bartelle, B. B. (2026, August). *Machine Vision-Enabled Laboratory Automation for Scalable Organoid Culture.* 2026 Beckman Symposium, Irvine, California, USA.

Crane, R. D.*, Alorf, A., Chen, Y. (2026, July). *A Memristive Synaptic Circuit Primitive to Establish Interaction Between Two Compact Subthreshold Izhikevich Silicon Neurons.* 2026 NSF Athena AI Institute Summer REU Showcase, Durham, North Carolina, USA.

Crane, R. D.*, Alorf, A., Chen, Y. (2026, July). *A Memristive Synaptic Circuit Primitive to Establish Interaction Between Two Compact Subthreshold Izhikevich Silicon Neurons.* 2026 Duke Summer Research Showcase, Durham, North Carolina, USA.

Crane, R. D.*, Bartelle, B. B. (2025, November). *Open Source Microscopy for Scalable Tissue Culture.* 2025 Arizona Regenerative Medicine Symposium, Phoenix, Arizona, USA.

Crane R. D.*, Bartelle, B. B. (2025, November). *Open Source Microscopy for Scalable Tissue Culture.* 2025 Fulton Forge Research Expo, Tempe, Arizona, USA.

Crane R. D.*, Kim, M.*, Salazar, N.*, Bartelle, B. B., et al. (2025, May). *Directed Evolution of alpha-galactosidase for Use in Clinical Xenotransfusions.* Biomedical Engineering Capstone Symposium, Tempe, Arizona, USA.

Crane R. D.*, Kim, M.*, Bartelle, B. B., et al. (2025, April). *Directed Evolution of alpha-galactosidase for Use in Clinical Xenotransfusions.* School of Life Sciences Undergraduate Research Symposium, Tempe, Arizona, USA.

ACADEMIC PROJECTS

AI Assistant for Defect and Fracture Analysis of Ti64 Titanium in Aerospace Manufacturing January 2026 - May 2026

Collaborated in a team to develop an AI assistant to guide decision making for titanium defect and image fracture analysis

- Trained a segmentation algorithm on Ti64 image defects and masks, and integrated the model into an LLM-based diagnosis interface and framework

Finite Element Analysis for Non-Invasive Mechanical Modeling of Intestinal Organoids January 2026 - May 2026

Independently replicated FEM of intestinal organoid with open-source tools

- Computed stress and strain fields specified with loads representing osmotic pressure and active contraction in FEniCS

Differences in Aromatherapy on Anxiety Between Sexes January 2025 - May 2025

Collaborated in a team to investigate physiological responses of aromatherapy between sexes

- Collected physiological response data using EKG, blood pressure cuffs, and pulse oximeter
- Analyzed collected data in Python

ASL-Translating Glove for Deaf and Speech-Impaired September 2024 – November 2024

Collaborated in a team to design glove for deaf and speech-impaired in under-resourced communities (Arduino IDE and ESP32):

- Designed an initial glove prototype using DIY flex sensors, made from copper tape using an ESP32 microcontroller
- Wrote programs in Arduino IDE for initial testing of handshapes to individual letters

Autonomous Device for Pest Removal August 2023 - November 2023

Led a team of four to design a pest removal device for homes and businesses in Arizona (Arduino):

- Managed project planning, troubleshooting, and team coordination from prototyping through final presentation
- Strengthened communication skills, adaptability, and collaborative problem-solving through hands-on prototyping and team coordination.

VOLUNTEER EXPERIENCE

- Volunteered at Grace Community Church, making coffee and serving it to attendees. June 2021 - July 2023
- Volunteered at Flagstaff Family Food Center and Hope Cottage Flagstaff Homeless Shelter, assisting in renovation of the food bank, as well as in preparing food and supplies, serving thousands of homeless men, women, and children

August 2021 - June 2022

ACTIVITIES

Visionary Hands
Engineering Projects in Community Service, Arizona State University September 2024 - November 2024
Finance Lead

- Developed team budget, corresponded with stakeholders including quality measurement for each major milestone (Microsoft Office)
- Pitched our project at the ASU EPICS Elite Pitch Competition 2024, placing third and receiving $1000 in prize money.

DIYBio
Synthetic Biology Club, Arizona State University August 2023 – Present
Member

- Actively participated in DIYBio's "Open Lab" series, where I gained hands-on experience in synthetic and molecular biology techniques, including PCR, gel electrophoresis, digestion, ligation, transformation, miniprep, and Golden Gate Assembly, fostering practical skills in a collaborative, educational setting.
- Taught undergraduates foundational circuit soldering skills

REFERENCES

Benjamin B. Bartelle, Assistant Professor
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Miyeko D. Mana, Assistant Professor
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